

MCQ Lab 2 - DSA in Java (Stack-Based)

All answers verified by compiling and executing each Java code snippet.

Question 1

What is the output of the following Java code?

Java Code:

```
public class Main {
    public static int evaluatePostfix(String expr) {
        int[] stack = new int[expr.length()];
        int top = -1;
        String[] tokens = expr.split(" ");
        for (String t : tokens) {
            if (t.matches("-?\\d+")) {
                stack[++top] = Integer.parseInt(t);
            } else {
                int b = stack[top--];
                int a = stack[top--];
                switch (t) {
                    case "+": stack[++top] = a + b; break;
                    case "-": stack[++top] = a - b; break;
                    case "*": stack[++top] = a * b; break;
                    case "/": stack[++top] = a / b; break;
                }
            }
        }
        return stack[top];
    }
    public static void main(String[] args) {
        System.out.println(evaluatePostfix("3 4 + 2 *"));
    }
}
```

Options: A) 14 B) 10 C) 16 D) 12

Compiled Output: 14

Correct Answer: A) 14

The postfix expression '3 4 + 2 *' is evaluated as follows:

- Push 3, Push 4 -> stack: [3, 4]
- '+' -> pop 4 and 3, compute 3+4=7, push 7 -> stack: [7]
- Push 2 -> stack: [7, 2]
- '*' -> pop 2 and 7, compute 7*2=14, push 14 -> stack: [14]
- Return 14.

Question 2

What is the output of the following Java code?

Java Code:

```
public class Main {
    public static void main(String[] args) {
        int[] stack = new int[5];
        int top = -1;
        for (int i = 1; i <= 4; i++) {
            stack[++top] = i * 10;
        }
        int sum = 0;
        while (top >= 0) {
            sum += stack[top--];
        }
    }
}
```

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```
    }  
    System.out.println(sum);  
}  
}
```

Options: A) 10 B) 100 C) 60 D) 40

Compiled Output: 100

Correct Answer: B) 100

The loop pushes 10, 20, 30, 40 onto the stack (i=1 to 4, each multiplied by 10). The while loop pops all elements and sums them: $40+30+20+10 = 100$.

Question 3

What is the output of the following Java code?

Java Code:

```
public class Main {  
    public static void main(String[] args) {  
        int[] stack = new int[5];  
        int top = -1;  
        stack[++top] = 100;  
        stack[++top] = 200;  
        top--;  
        top--;  
        System.out.println(top);  
    }  
}
```

Options: A) -1 B) 0 C) 100 D) ArrayIndexOutOfBoundsException

Compiled Output: -1

Correct Answer: A) -1

Initially $top = -1$. After $stack[++top] = 100$, top becomes 0. After $stack[++top] = 200$, top becomes 1. Then $top--$ makes it 0, and another $top--$ makes it -1. No array access happens at index -1, so no exception is thrown.

Question 4

What is the output of the following Java code?

Java Code:

```
public class Main {  
    public static void main(String[] args) {  
        int[] stack = new int[10];  
        int top = -1;  
        stack[++top] = 3;  
        stack[++top] = 7;  
        stack[++top] = 1;  
        stack[++top] = 9;  
        int max = stack[0];  
        for (int i = 1; i <= top; i++) {  
            if (stack[i] > max) max = stack[i];  
        }  
        System.out.println(max);  
    }  
}
```

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```
}
```

Options: A) 3 B) 7 C) 1 D) 9

Compiled Output: 9

Correct Answer: D) 9

The stack contains [3, 7, 1, 9] with top = 3. The loop finds the maximum value starting from max = stack[0] = 3. Comparing: 7>3 -> max=7, 1<7 -> no change, 9>7 -> max=9.

Question 5

What is the output of the following Java code?

Java Code:

```
public class Main {
    public static int evaluatePrefix(String expr) {
        int[] stack = new int[expr.length()];
        int top = -1;
        String[] tokens = expr.split(" ");
        for (int i = tokens.length - 1; i >= 0; i--) {
            String t = tokens[i];
            if (t.matches("-?\\d+")) {
                stack[++top] = Integer.parseInt(t);
            } else {
                int a = stack[top--];
                int b = stack[top--];
                switch (t) {
                    case "+": stack[++top] = a + b; break;
                    case "-": stack[++top] = a - b; break;
                    case "*": stack[++top] = a * b; break;
                    case "/": stack[++top] = a / b; break;
                }
            }
        }
        return stack[top];
    }
    public static void main(String[] args) {
        System.out.println(evaluatePrefix("* - 5 3 2"));
    }
}
```

Options: A) 10 B) 12 C) 6 D) 4

Compiled Output: 4

Correct Answer: D) 4

The prefix expression '* - 5 3 2' is evaluated right-to-left:

- Process tokens right to left: 2 -> push; 3 -> push; 5 -> push; '-' -> pop 5 and 3, compute 5-3=2, push 2; '*' -> pop 2 and 2, compute 2*2=4, push 4.
- Return 4.

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All answers verified by compiling and executing each Java code snippet.

Question 6

What is the output of the following Java code?

Java Code:

```
public class Main {
    public static void main(String[] args) {
        int[] stack = new int[5];
        int top = -1;
        stack[++top] = 5;
        stack[++top] = 15;
        stack[++top] = 25;
        int popped = stack[top--];
        System.out.println(popped + " " + top);
    }
}
```

Options: A) 25 2 B) 25 1 C) 15 0 D) 15 1

Compiled Output: 25 1

Correct Answer: B) 25 1

After pushing: stack = [5, 15, 25], top = 2. 'popped = stack[top--]' reads stack[2]=25 first, then decrements top to 1. Output: 25 1.

Question 7

What is the output of the following Java code?

Java Code:

```
public class Main {
    public static int precedence(char c) {
        if (c == '+' || c == '-') return 1;
        if (c == '*' || c == '/') return 2;
        return 0;
    }
    public static String infixToPostfix(String expr) {
        char[] stack = new char[expr.length()];
        int top = -1;
        String result = "";
        for (char c : expr.toCharArray()) {
            if (Character.isLetterOrDigit(c)) {
                result += c;
            } else if (c == '(') {
                stack[++top] = c;
            } else if (c == ')') {
                while (top >= 0 && stack[top] != '(') {
                    result += stack[top--];
                }
                top--;
            } else {
                while (top >= 0 && precedence(stack[top]) >= precedence(c)) {
                    result += stack[top--];
                }
                stack[++top] = c;
            }
        }
        while (top >= 0) {
            result += stack[top--];
        }
    }
}
```

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All answers verified by compiling and executing each Java code snippet.

```
    }
    return result;
}
public static void main(String[] args) {
    System.out.println(infixToPostfix("A+B*C"));
}
}
```

Options: A) ABC*+ B) ABC+* C) AB+C* D) A+BC*

Compiled Output: ABC*+

Correct Answer: A) ABC*+

For 'A+B*C':

- A -> output: A
- '+' -> push to stack
- B -> output: AB
- '*' -> precedence 2 > 1, push to stack
- C -> output: ABC
- End: pop '*' -> output: ABC*, pop '+' -> output: ABC*+
- Result: ABC*+

Question 8

What does the following method return when called with the input "{[()]}"?

Java Code:

```
public class Main {
    public static boolean isBalanced(String expr) {
        char[] stack = new char[expr.length()];
        int top = -1;
        for (char c : expr.toCharArray()) {
            if (c == '(' || c == '[' || c == '{') {
                stack[++top] = c;
            } else if (c == ')' && top >= 0 && stack[top] == '(') {
                top--;
            } else if (c == ']' && top >= 0 && stack[top] == '[') {
                top--;
            } else if (c == '}' && top >= 0 && stack[top] == '{') {
                top--;
            } else {
                return false;
            }
        }
        return top == -1;
    }
    public static void main(String[] args) {
        System.out.println(isBalanced("{[()]}"));
    }
}
```

Options: A) false B) Compilation Error C) 0 D) true

Compiled Output: true

Correct Answer: D) true

The string '{[()]}' is perfectly balanced. Each closing bracket matches the most recent opening bracket: '{' -> '[' -> '(' -> ')' pops '(' -> ']' pops '[' -> '}' pops '{'. Stack is empty, top == -1, returns true.

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All answers verified by compiling and executing each Java code snippet.

Question 9

What is the output of the following Java code?

Java Code:

```
public class Main {
    public static boolean isBalanced(String expr) {
        char[] stack = new char[expr.length()];
        int top = -1;
        for (char c : expr.toCharArray()) {
            if (c == '(' || c == '[' || c == '{') {
                stack[++top] = c;
            } else if (c == ')' && top >= 0 && stack[top] == '(') {
                top--;
            } else if (c == ']' && top >= 0 && stack[top] == '[') {
                top--;
            } else if (c == '}' && top >= 0 && stack[top] == '{') {
                top--;
            } else {
                return false;
            }
        }
        return top == -1;
    }
    public static void main(String[] args) {
        System.out.println(isBalanced("{[]}"));
    }
}
```

Options: A) false B) true C) Compilation Error D) 1

Compiled Output: false

Correct Answer: A) false

For '{[]}': '(' pushed, '{' pushed, '[' pushed. Then ')' is encountered but stack[top] = '[' which is not '('. The condition stack[top] == '(' fails, so it falls to the else block and returns false.

Question 10

What is the output of the following Java code?

Java Code:

```
public class Main {
    public static boolean isBalanced(String expr) {
        char[] stack = new char[expr.length()];
        int top = -1;
        for (char c : expr.toCharArray()) {
            if (c == '(' || c == '[' || c == '{') {
                stack[++top] = c;
            } else if (c == ')' && top >= 0 && stack[top] == '(') {
                top--;
            } else if (c == ']' && top >= 0 && stack[top] == '[') {
                top--;
            } else if (c == '}' && top >= 0 && stack[top] == '{') {
                top--;
            } else {
                return false;
            }
        }
    }
}
```

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All answers verified by compiling and executing each Java code snippet.

```
        return top == -1;
    }
    public static void main(String[] args) {
        System.out.println(isBalanced("(((("));
    }
}
```

Options: A) false B) Compilation Error C) true D) 1

Compiled Output: false

Correct Answer: A) false

For '((((': Three '(' are pushed (top=0,1,2). Then ')' matches stack[top]='(', so top becomes 1. After the loop, top = 1 which is not -1, so top == -1 is false. Returns false.

Question 11

What is the output of the following Java code?

Java Code:

```
public class Main {
    public static int evaluatePostfix(String expr) {
        int[] stack = new int[expr.length()];
        int top = -1;
        String[] tokens = expr.split(" ");
        for (String t : tokens) {
            if (t.matches("-?\\d+")) {
                stack[++top] = Integer.parseInt(t);
            } else {
                int b = stack[top--];
                int a = stack[top--];
                switch (t) {
                    case "+": stack[++top] = a + b; break;
                    case "-": stack[++top] = a - b; break;
                    case "*": stack[++top] = a * b; break;
                    case "/": stack[++top] = a / b; break;
                }
            }
        }
        return stack[top];
    }
    public static void main(String[] args) {
        System.out.println(evaluatePostfix("3 4 + 2 *"));
    }
}
```

Options: A) 14 B) 9 C) 24 D) 10

Compiled Output: 14

Correct Answer: A) 14

The postfix expression '3 4 + 2 *' evaluates as: Push 3, Push 4, '+' gives 7, Push 2, '*' gives 7*2 = 14.

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All answers verified by compiling and executing each Java code snippet.

Question 12

What is the output of the following Java code?

Java Code:

```
public class Main {
    public static void main(String[] args) {
        char[] stack = new char[10];
        int top = -1;
        String s = "HELLO";
        for (int i = 0; i < s.length(); i++) {
            stack[++top] = s.charAt(i);
        }
        String result = "";
        while (top >= 0) {
            result += stack[top--];
        }
        System.out.println(result);
    }
}
```

Options: A) OLELH B) OLLEH C) LOLEH D) HELLO

Compiled Output: OLLEH

Correct Answer: B) OLLEH

The string "HELLO" is pushed character by character onto the stack (H, E, L, L, O). Then characters are popped in reverse order: O, L, L, E, H. Result: OLLEH.

Question 13

What is the output of the following Java code?

Java Code:

```
public class Main {
    public static int precedence(char c) {
        if (c == '+' || c == '-') return 1;
        if (c == '*' || c == '/') return 2;
        return 0;
    }
    public static String infixToPostfix(String expr) {
        char[] stack = new char[expr.length()];
        int top = -1;
        String result = "";
        for (char c : expr.toCharArray()) {
            if (Character.isLetterOrDigit(c)) {
                result += c;
            } else if (c == '(') {
                stack[++top] = c;
            } else if (c == ')') {
                while (top >= 0 && stack[top] != '(') {
                    result += stack[top--];
                }
                top--;
            } else {
                while (top >= 0 && precedence(stack[top]) >= precedence(c)) {
                    result += stack[top--];
                }
                stack[++top] = c;
            }
        }
        while (top >= 0) {
            result += stack[top--];
        }
    }
}
```


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```
    }
    }
    while (top >= 0) {
        result += stack[top--];
    }
    return result;
}
public static void main(String[] args) {
    System.out.println(infixToPostfix("(A+B)*(C-D)"));
}
}
```

Options: A) AB+*CD- B) A+B*C-D C) ABCD+-* D) AB+CD-*

Compiled Output: AB+CD-*

Correct Answer: D) AB+CD-*

For '(A+B)*(C-D)':

- '(' -> push, A -> output: A, '+' -> push, B -> output: AB, ')' -> pop '+', output: AB+
- '*' -> push
- '(' -> push, C -> output: AB+C, '-' -> push, D -> output: AB+CD, ')' -> pop '-', output: AB+CD-
- End: pop '*', output: AB+CD-*
- Result: AB+CD-*

Question 14

What is the output of the following Java code?

Java Code:

```
public class Main {
    public static void main(String[] args) {
        int[] stack = new int[5];
        int top = -1;
        stack[++top] = 10;
        stack[++top] = 20;
        stack[++top] = 30;
        top--;
        System.out.println(stack[top]);
    }
}
```

Options: A) Compilation Error B) 30 C) 20 D) 10

Compiled Output: 20

Correct Answer: C) 20

After pushing: stack = [10, 20, 30], top = 2. The 'top--' decrements top to 1. Then stack[top] = stack[1] = 20.

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All answers verified by compiling and executing each Java code snippet.

Question 15

What is the output of the following Java code?

Java Code:

```
public class Main {
    public static int evaluatePrefix(String expr) {
        int[] stack = new int[expr.length()];
        int top = -1;
        String[] tokens = expr.split(" ");
        for (int i = tokens.length - 1; i >= 0; i--) {
            String t = tokens[i];
            if (t.matches("-?\\d+")) {
                stack[++top] = Integer.parseInt(t);
            } else {
                int a = stack[top--];
                int b = stack[top--];
                switch (t) {
                    case "+": stack[++top] = a + b; break;
                    case "-": stack[++top] = a - b; break;
                    case "*": stack[++top] = a * b; break;
                    case "/": stack[++top] = a / b; break;
                }
            }
        }
        return stack[top];
    }
    public static void main(String[] args) {
        System.out.println(evaluatePrefix("* + 2 3 4"));
    }
}
```

Options: A) 24 B) 20 C) 14 D) 10

Compiled Output: 20

Correct Answer: B) 20

The prefix expression '* + 2 3 4' is evaluated right-to-left:

- 4 -> push; 3 -> push; 2 -> push; '+' -> pop 2 and 3, compute 2+3=5, push 5; '*' -> pop 5 and 4, compute 5*4=20, push 20.
- Return 20.